

CUSTOMER SEGMENTATION ANALYSIS USING PYTHON AND SQL

ABSTRACT

Businesses generate large volumes of customer data through transactions and interactions. Analysing customer behaviour helps organizations understand purchasing patterns, identify valuable customers, and improve marketing strategies. Without proper analysis, businesses may struggle to target the right customers and maximize revenue.

This project focuses on Customer Segmentation Analysis using Python and SQL. The Mall Customers dataset was analysed to study customer demographics, spending behaviour, and purchasing patterns. Data cleaning, exploratory data analysis, customer profiling, and segmentation techniques were applied to identify high-value customers and generate business insights.

Various visualizations including Age Group Analysis, Gender Analysis, Customer Segment Distribution, and Correlation Heatmap were developed. The project helps businesses understand customer behaviour and supports data-driven marketing and customer retention strategies.

CHAPTER 1: INTRODUCTION

1.1 Background

Customer Segmentation is the process of dividing customers into groups based on common characteristics such as age, income, gender, and spending behaviour. It helps organizations understand customer needs and design effective marketing strategies.

Modern businesses collect large amounts of customer data. Analysing this data enables organizations to identify profitable customer groups, improve customer satisfaction, and increase business growth.

1.2 Problem Statement

Businesses often struggle to identify high-value customers and understand spending patterns. Manual analysis of customer data is time-consuming and may not provide actionable insights. Therefore, customer segmentation is required to classify customers and support business decision-making.

1.3 Objectives

- Analyse customer demographics and spending behaviour.
- Identify high-value customers.
- Perform customer segmentation.
- Study age-wise and gender-wise spending patterns.
- Generate business insights through visualizations.
- Apply SQL queries for customer data analysis.

1.4 Scope of the Project

The project can be used by:

- Marketing Teams

- Business Analysts
- Data Analysts
- Customer Relationship Managers
- Sales Teams

to understand customer behaviour and improve marketing effectiveness.

CHAPTER 2: PYTHON AND SQL OVERVIEW

2.1 Introduction to Python

Python is a powerful programming language widely used in data analysis, machine learning, and business analytics. Python provides libraries for data cleaning, visualization, and statistical analysis.

2.2 Introduction to SQL

Structured Query Language (SQL) is used to store, retrieve, and analyse data from databases. SQL enables users to perform aggregations, filtering, joins, and analytical operations efficiently.

2.3 Tools Used

Python

Used for data cleaning, exploratory data analysis, segmentation, and visualization.

Jupyter Notebook

Used for executing Python code and performing analysis.

SQL Plus

Used for SQL query execution and analytical operations.

Pandas

Used for data manipulation and cleaning.

Matplotlib and Seaborn

Used for data visualization.

2.4 Advantages

- Efficient customer analysis
- Better marketing strategies
- Improved customer targeting
- Business insight generation
- Data-driven decision-making

CHAPTER 3: LITERATURE SURVEY

Study 1

Customer segmentation improves marketing effectiveness by grouping customers with similar characteristics and purchasing behaviour.

Study 2

Spending behaviour analysis helps businesses identify profitable customers and optimize marketing campaigns.

Study 3

Data visualization techniques help analysts understand customer trends and patterns more effectively.

Study 4

Customer profiling enables organizations to personalize services and improve customer retention.

CHAPTER 4: METHODOLOGY

4.1 Dataset Description

The Mall Customers dataset was used for customer segmentation analysis.

Dataset Attributes

- CustomerID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)

4.2 Data Import

Steps:

1. Download Mall Customers dataset.
2. Import dataset into Jupyter Notebook.
3. Verify data structure.
4. Load dataset using Pandas.

4.3 Data Cleaning

The dataset was checked for:

- Missing values
- Duplicate records
- Data consistency

- Data quality validation

Results:

- Missing Values = 0
- Duplicate Records = 0

4.4 Exploratory Data Analysis

The following analyses were performed:

- Age Group Analysis
- Gender Analysis
- High-Value Customer Identification
- Customer Segmentation
- Correlation Analysis

4.5 Visualizations Created

- Average Spending Score by Age Group
- Average Spending Score by Gender
- Customer Segment Distribution
- Correlation Heatmap

CHAPTER 5: SYSTEM ARCHITECTURE

Mall Customers Dataset



Data Import



Data Cleaning



Exploratory Data Analysis



Customer Segmentation



Correlation Analysis



Visualization



Business Insights

CHAPTER 6: RESULTS AND ANALYSIS

6.1 Dataset Summary

Metric Value

Total Customers 200

Average Age 38.85

Average Annual Income 60.56 k\$

Average Spending Score 50.20

6.2 Age Group Analysis

Observation:

- Age group 26–35 has the highest average spending score (64.45).
- Younger customers tend to spend more than older customers.

6.3 Gender Analysis

Observation:

- Female customers have an average spending score of 51.53.
- Male customers have an average spending score of 48.51.
- Female customers spend slightly more than male customers.

6.4 Customer Segmentation Analysis

Observation:

- Medium Value Customers = 83
- Low Value Customers = 63

- High Value Customers = 54

6.5 Correlation Analysis

Observation:

- Age and Spending Score show a negative correlation (-0.327).
- Annual Income has almost no relationship with Spending Score.
- Younger customers tend to have higher spending scores.

CHAPTER 7: INFERENCE AND DISCUSSION

Key Findings

1. Dataset contains 200 customers.
2. Age group 26–35 is the highest spending customer segment.
3. Female customers spend slightly more than male customers.
4. 54 high-value customers were identified.
5. Medium-value customers form the largest customer group.
6. Age influences spending behaviour more than income.

Benefits of the Project

- Improved customer understanding
- Customer behaviour analysis
- High-value customer identification
- Better marketing strategies
- Data-driven decision-making

CHAPTER 8: FUTURE SCOPE

Future enhancements may include:

- Real-time customer transaction analysis
- Machine Learning-based customer segmentation
- Customer churn prediction
- Recommendation systems
- SQL database integration
- Interactive Power BI dashboard development

CONCLUSION

The Customer Segmentation Analysis project successfully analysed customer demographics and spending behaviour using Python and SQL. Data cleaning, exploratory data analysis, customer segmentation, and correlation analysis were performed to generate meaningful business insights.

The project identified high-value customers, spending trends, and customer segments that can help organizations improve marketing strategies and customer retention. The project demonstrates the effectiveness of Python and SQL in customer analytics and business intelligence.

REFERENCES

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3. Matplotlib Documentation
4. Seaborn Documentation
5. Oracle SQL Documentation
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